UNIVERSITY OF KELANIYA
DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR 2022/2023
BACHELOR OF SCIENCE DEGREE
BECS 22623 - IMPLEMENTATION OF NUMERICAL METHODS USING APPROPRIATE SOFTWARE

Semester II | Tutorial - 02

1. The butterfly curve is given by the following parametric equations:

$$x = sin(t)(e^{cos(t)} - 2cos(4t) - sin^5(t/12))$$
$$y = cos(t)(e^{cos(t)} - 2cos(4t) - sin^5(t/12))$$

Create a vector $t$ with values from $0$ to $20$ with an increment of $0.01$. Create subplots in a 2 x 1 arrangement (stacked vertically) with

- The top panel plots the position (both $x$ and $y$) against $t$.
- The bottom panel plotting $y$ against $x$, as a magenta line.
- Title the bottom subplot as "The Butterfly".
- Label the axes appropriately.

2. The capacitance of two parallel conductors of length $L$ and radius $r$, separated by distance $d$ in air, is given by

$$C = \frac{\pi \epsilon L}{\ln(\frac{d-r}{r})}$$

Where $\epsilon$ is the permittivity of air ($\epsilon = 8.854$ x $10^{-12} F/m$).

Write a script that accepts user input for $d, L$ and $r$ and computes and displays $C$. Test the script file with the values $L = 1m, r = 0.001m$, and $d = 0.004m$.

3. The blades of a wind turbine ($12m$ long) have an initial angular velocity of $6RPM$, and a wind gust blows from $t = 0$ to $t = 5s$. The blades undergo an angular acceleration of $\alpha = Cw\ rad/s^2$ , where $C$ is a constant and $w$ is the speed of the gust in $m/s$.

(a) Assume that the wind gust speed is a constant $4m/s$ during this period and that $C = 0.04$ $rad.m^{-1}.s^{-1}$ . Plot the translational speed at the tip of the blades over $t = [0, 5]$. On a second plot, plot the radial acceleration over this time. Label plots appropriately.

(b) Write a function file which can calculate the tip speed of this wind turbine for a wind gust of arbitrary speed and duration, for an arbitrary initial angular velocity.

Figure 1: Mannar Wind Energy Park

4. Write a program in a script file that determines the real roots of a quadratic equation $ax^2+bx+c=0$. Name the file "quadroots". When the file runs, it asks the user to enter the values of constants $a, b$ and $c$. To calculate roots of the equation the program calculates the discriminant $D$, given by:

$$D = b^2 - 4ac$$

If $D > 0$,

The program displays the message "The equation has two distinct real roots." and the roots are displayed in the next line.

If $D = 0$,
The program displays the message "The equation has one roots." and the roots are displayed in the next line.

If $D < 0$,
The program displays the massage "The equation has no real roots."

Run the script file in the Command window three times to obtain solutions to the following three equations.

(a) $-6x^2 + 11x + 35 = 0$
(b) $3x^2 + 4x + 2 = 0$
(c) $2x^2 + 8x + 8 = 0$

. . . . . .